

CAO XIZHEN

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ABOUT ME

JC2 student from Hwa Chong Institution, currently a member of Hwa Chong Infocomm & Robotics Society, President of Project Small Bytes(HCI branch), and has been enrolled in the MOE Art Elective Programme since 2021. My main areas of interest consist of Mathematics, Science Research, and Machine Learning, as well as cybersecurity, computer science, and art.

ACHIEVEMENTS/COMPETITIONS

2026

- ★ Bronze, National Olympiad in Informatics (NOI) 2026
- ★ Honourable Mention, National Cybersecurity Olympiad (NCO) 2026
- ★ Finalist, National Olympiad in Artificial Intelligence (NOAI) 2026
- ★ Overall Champion + Winner of AI Challenge, Cyberthon 2026
- ★ Finalist, Singapore Science & Engineering Festival (SSEF) 2026
- ★ 1st Runner Up, Lag and Crash CTF 2026

2025

- ★ 1st Place, AI CTF 2025 Qualifiers, 4th Place Finals
- ★ 6th Place, Singapore Informatics League (SIL)
- ★ Finalist, DSTA Brainhack CDDC 2025
- ★ Finalist, GreyCTF 2025
- ★ 8th Place Lag and Crash CTF 2025
- ★ 3rd Place NES Connect 2025
- ★ 2nd Runner Up, Articulation Prize 2025

2024

- ★ Winner of JP Morgan GenerationTech 2024
- ★ Bronze, Singapore Mathematics Project Festival 2024
- ★ Finalist, National Olympiad in Informatics 2024
- ★ Awarded Lively Places Fund under Urban Redevelopment Authority (URA)
- ★ Honourable Mention, SMO Open 2024
- ★ Recipient of Nanyang Honour Roll

2023

- ★ Honourable Mention, SMO Open & Senior 2023
- ★ Distinction, Mathematics research project in Number Theory & Cryptography

EXPERIENCES/PROJECTS/EVENTS

★ Research

- Research Intern @ A*STAR IMRE
 - Project title: Implementation of quantum logic gates with solid-state qubits interfaced with an optical metasurface
- Student researcher @ Centre for Quantum Technologies (CQT) & NUS

- Project title: [Optimising Bell Measurement Settings in Entanglement-Based Quantum Key Distribution Under Finite Key Conditions](#)
 - Technical paper to be presented at IEEE-Smartnets Conference 2026
 - Finalist project for SSEF 2026
 - List III: Project on cryptanalysis techniques & number theory in 2023 & 2024
 - Project on probability & combinatorics, developed an algorithm to solve Wordle in 2022
- ★ **SiebrsecCTF (SCTF) 2025 & 2026**
 - Core organising team member for SCTF, a CTF with ~1,000 participants (2026)
 - Publicity Head, Ceremony IC, challenge setter and planner for both years.
- ★ **Community service/volunteering**
 - Project Small Bytes (2025-2026)
 - Teaching primary 5 level schoolchildren Python programming and Pygame
 - Singapore Red Cross (member)
- ★ **Conferences/events**
 - DEF CON 2026
 - International Science Youth Forum 2026 (Facilitator)
 - Wavesparks Future Re:Wired @ SWITCH 2025 (Coordinator)
 - Off-By-One Conference 2025
 - SuperAI Conference 2025
- ★ **Sentinel Programme**
 - Sentinel Pilot Programme from 2022 - 2023
 - Awarded Certificate of Achievement for 2 years
- ★ **Art Elective Programme (AEP)**
 - HCI Art Elective Programme (PEACH) 2025-2026
 - NYGH Art Elective Programme 2021-2024

SKILLS & INTERESTS

- ★ **Game Development/Design**
 - Proficient in Unity, Godot, Blender
 - Skilled in 3D modelling, storyboarding, animation
- ★ **Programming languages**
 - C#, C++, Python
 - Javascript, HTML, CSS
- ★ **Cybersecurity**
 - Cryptography
 - Advanced modular arithmetic, group theory
 - Familiar with cryptosystems such as RSA, ECC and AES
 - Binary Exploitation (Pwn)
 - Experience with tools such as pwndbg and Ghidra
 - Familiar with Stack-based binary exploitation
- ★ **Competitive Programming / Data Structures & Algorithms**
 - Familiar with various medium-level algorithms & competitive programming skills
- ★ **Quantum Computing**
 - Familiar with various Quantum Key Distribution protocols
 - Experience in software and Python libraries such as Qiskit, Netquid, and QuTiP.